

PUSHKAR SINGH

Bangalore, India · Open to Visa Sponsorship

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Summary

AI/ML Engineer with 2+ years of experience building agentic and machine learning solutions for banking and financial services workflows. Skilled in Python, data analysis, and deploying scalable AI systems using RAG and LangGraph. Experienced in model evaluation, experimentation, and translating business problems into data-driven solutions in consulting environments.

Experience

PwC - PricewaterhouseCoopers

Feb. 2024 – Present

Associate 2 (Gen AI)

• ACTA - Agentic Controls Testing Assistant

- Designed and implemented ACTA as a multi-agent LangGraph-based system, orchestrating parallel agents using RAG & semantic indexing to analyse structured & unstructured financial datasets, identify patterns & automate audit workflows.
- Built and deployed on Azure using LangChain, Azure OpenAI, and Microsoft Graph connectors to extract, clean, and transform enterprise financial data, leveraging embeddings, OCR, and multimodal pipelines to generate structured audit insights and reduce manual effort by 60%.
- Implemented LLM-as-a-judge and cross-encoder based confidence scoring to measure model performance using evaluation metrics, analyse errors & improve accuracy by 40%, while optimizing latency through chunking and retrieval tuning.

• CLUTS - Custom GPT based solution

- Built CLUTS, a GPT-powered pipeline with multi-step reasoning using prompt engineering and RAG to process and normalize large financial datasets, detect anomalies, and generate structured outputs, adapting solutions across 5+ client engagements.
- Deployed on Azure using LangChain and retrieval pipelines to analyse financial control data, generate test strategies, and optimize workflows, enabling scalable data-driven decision-making and improving audit efficiency.

Insignia Consultancy Solutions

May 2023 – Aug. 2023

Microservices Developer Intern

- Collaborated with development team to design and implement Java SpringBoot microservices for subscription management, including input validation and transaction rollback.
- Developed REST endpoints, creating JPA entities and optimizing MySQL queries for accuracy and performance.

Education

Graphic Era Hill University, Dehradun

Sep. 2020 – Jun. 2024

B.Tech. in Computer Science & Engineering (8.25/10)

Maria Assumpta Sr. Sec. Convent School, Kashipur

Mar. 2019 – Apr. 2020

Senior Secondary Education - XII (85%)

Skills

- **AI Systems & tools:** Generative AI, Agentic AI, RAG, LangChain, LLMs, MCP Servers, Multi-Agent system
- **Languages & Frameworks:** Python (Pandas, NumPy), Java, FastAPI, SpringBoot Microservices, REST APIs
- **Data & Analytics:** SQL, Data Preprocessing, Feature Engineering, Exploratory Data Analysis, Model Evaluation
- **Cloud, DevOps & Databases:** Azure OpenAI, Docker, MLOps, Git, DevOps, ChromaDB, MongoDB
- **Professional Skills:** Problem Solving, Adaptability, Client Management, Structured Communication, Active Listening

Projects

Roxy - Interview Voice Assistant (Llama, FastAPI, RAG, LangGraph, ChromaDB, Whisper, Coqui)

- Designed architecture for Roxy using LangGraph as a multimodal system with orchestrated agents handling content analysis, question generation, and evaluation, enabling real-time processing of structured and unstructured datasets.
- Built and deployed using FastAPI, ChromaDB, Whisper, and Coqui TTS, leveraging RAG, embeddings, and data preprocessing techniques to extract insights from PDFs, audio, and video for retrieval-backed responses.
- Implemented a feedback-driven pipeline to analyse model outputs, perform iterative improvements, and optimize system performance using evaluation metrics and workflow-level experimentation.

Certifications & Publications

- Selected for **McKinsey Forward Program**, focused on problem solving, digital, and leadership skills.
- **AWS Certified Cloud Practitioner**
- **Published research** on breakout/reversal trading using candlestick-close and RSI divergence strategies. **IEEE Xplore**
- Awarded **PwC's Extra Miler Award & Quarterly LoS** for taking Agentic AI initiatives and project deliverables.